

DOMINIK KLUMPP (Dr. rer. nat.)

PERSONAL INFORMATION

Born in São Paulo, Brazil, 1994

nationality German, Brazilian

email dominik.klumpp@lix.polytechnique.fr

website dominik-klumpp.net

EDUCATION

PhD studies
(Dr. rer. nat)
Computer Science

2019–2025 University of Freiburg

PhD research at the chair for Software Engineering, centered around automatic program verification, in particular for concurrent programs. Worked as a lecturer and teaching assistant for several lectures at the Chair of Software Engineering. Supervised 20 student projects, B.Sc. and M.Sc. theses.

Thesis: *From Commutativity to Proofs and Back Again:
Reduction and Verification of Concurrent Programs*

My thesis investigates the effect of *commutativity reasoning* (and *partial order reduction techniques*) on simplifying and speeding up the fully-automated, algorithmic verification of concurrent programs over infinite data types. I implemented the verification algorithms arising from my theoretical results in **ULTIMATE GEMCUTTER**, a state-of-the-art algorithmic verification tool for concurrent C programs.

Final grade: o (with distinction / *summa cum laude*)

Advisor: Prof. Andreas Podelski

M.Sc. Software
Engineering

2016–2018 University of Augsburg, TU Munich, LMU Munich

Elite Graduate Program *Software Engineering* by the University of Augsburg, the Technical University of Munich and the Ludwig-Maximilians-University Munich. The program is centered around the five subject areas *Software Engineering, Formal Methods, Distributed Systems, Databases* and *Human-Computer Interaction*. My personal focus was on the first two of these topics, especially on *Formal Methods*.

Thesis: *Automated Control Flow Reconstruction from Assembler Programs*

The research for my master thesis was conducted at and in cooperation with Macquarie University, Sydney. I adapted the automated verification technique *trace abstraction refinement* in order to build provably sound and precise control flow graphs for assembler programs with indirect branches.

Final grade: 1.0

[grading scheme: 1.0 (best) – 5.0 (failed)]

Advisor: Prof. Wolfgang Reif

Supervisors: Prof. Franck Cassez (Macquarie University), Dr. Gerhard Schellhorn

B.Sc. Computer
Science

2013–2016 University of Augsburg

Thesis: *Optimising Runtime Safety Analysis Efficiency for Self-Organising Systems*

I wrote my bachelor thesis at the *Institute for Software and Systems Engineering* at the University of Augsburg. A short version of the thesis was published at the **QA4SASO** workshop on the *FAS* 2016* conference in Augsburg.

Final grade: 1.3

[grading scheme: 1.0 (best) – 5.0 (failed)]

Advisor: Prof. Wolfgang Reif

Supervisors: Dr. Axel Habermeier, Dr. Benedikt Eberhardinger, Dr. Hella Seebach

WORK EXPERIENCE

PostDoc	since 06/2025	PostDoc Researcher at Laboratoire d'Informatique (LIX), CNRS – ÉCOLE POLYTECHNIQUE, France
		Research on automated and deductive verification of concurrent programs.
Researcher / PhD candidate	2019 - 2025	Researcher & PhD candidate at the Chair for Software Engineering, UNIVERSITY OF FREIBURG
		Research on automatic verification with a focus on concurrent programs. Lecturer for the lecture <i>Program Verification</i> in the summer semester 2024.
		Co-organised several lectures offered by the Software Engineering group, in particular <i>Program Verification</i> , <i>Cyber-Physical Systems: Discrete Models</i> and <i>Theoretical Computer Science</i> . This involved designing weekly exercise sheets, coordinating student tutors, marking students' submissions, teaching a weekly exercise session, occasionally substituting for the lecturer, as well as designing, administering and marking the final exam.
		Supervised students in several seminars offered by the Software Engineering group, in particular <i>Automata Theory</i> and <i>Program Analysis</i> . Supervised 20 B.Sc. and M.Sc. theses and projects.
Student Research Assistant	2016–2018	Student Research Assistant at the Institute for Software and Systems Engineering, UNIVERSITY OF AUGSBURG
		Research on safety analysis and on quality assurance for self-organizing systems. Contributed to the development of the S# modeling, safety analysis and simulation framework. Modeled and analyzed multiple case studies of self-organizing systems. Co-authored 3 scientific publications.
Zuehlke	Summer 2017	Internship, ZUEHLKE ENGINEERING AG — Munich
		Developed an Angular web application with a dockerized microservice backend based on Java and Spring Boot. Interfaced with several project and development management tools to display project status information.
Tutor: Theoretical Computer Science	2015–2017	Tutor at the Chair for Theoretical Computer Science, UNIVERSITY OF AUGSBURG
		Worked on the lecture <i>Introduction to Theoretical Computer Science</i> in the summer semesters 2015, 2016 and 2017. This involved teaching a weekly exercise class, marking students' homework and assisting in supervision and marking of the exam.
Tutor: Discrete Structures	2015–2016	Tutor at the Chair for Programming Methodology and Multimedia Information Systems, UNIVERSITY OF AUGSBURG
		Worked on the lecture <i>Discrete Structures for Computer Scientists</i> in the winter semesters 2015 and 2016. This involved teaching a weekly exercise class, marking students' homework and assisting in supervision and marking of the exam.
itestra	Summer 2015	itestra CodingCamp internship, ITESTRA — Munich
		Developed a prototype for a Windows mobile app with a Java backend.

ACADEMIC SERVICE

Reviewer	PLDI'26	Review Committee member
	SPIN'25	Program Committee member
	OOPSLA	External Reviewer ('22 & '23)
	SV-COMP	Program Committee member ('21-'26)
		Jury member for Ultimate Kojak ('21) Jury member for Ultimate GemCutter ('22 - '26)

Organiser	VDS'26	Organiser of the 5th Workshop on Verification of Distributed Systems (VDS'26), 8 th - 10 th June 2026 in Agadir, Morocco. netys.net/wds2026/VDS26.html
	AVM'24	Organised the 16th Alpine Verification Meeting (AVM'24), 4 th - 6 th September 2024 in Freiburg (67 participants). avm2024.informatik.uni-freiburg.de

INVITED TALKS & RESEARCH VISITS

Rabat, Morocco	May 2025	Workshop on Verification of Distributed Systems (VDS 2025) 3-day invitation-only workshop, colocated with the NETYS conference. Talk: Finding Commutativity for Algorithmic Verification of Concurrent Programs
Kaiserslautern, Germany	Mar 2025	Visit at Max-Planck-Institute for Software Systems Seminar talk and discussions about possible collaborations. Talk: How Commutativity Simplifies Proofs of Concurrent Programs
Paris, France	Feb 2025	Visit at Université Paris Cité (IRIF) and École Polytechnique (LIX) Seminar talk and discussions about possible collaborations. Talk: How Commutativity Simplifies Proofs of Concurrent Programs
Vienna, Austria	May 2024	Visit at IST Austria Seminar talk and discussions about possible collaborations. Talk: How Commutativity Simplifies Proofs of Concurrent Programs
Uppsala, Sweden	Feb 2024	Visit at Uppsala Universitet Collaboration on a joint project.
Toronto, Canada	Jun 2022	Visit at University of Toronto Seminar talk and collaboration on a joint paper. Talk: Sound Sequentialization for Concurrent Program Verification
Sydney, Australia	Apr-Oct 2018	Visiting researcher at Macquarie University 6-month research stay for Master thesis research.

AWARDS

University of Freiburg	SUMMER SEMESTER 2024	Teaching Award of the Faculty of Engineering for the best lecture in Computer Science (Cyber-Physical Systems II: Program Verification)
SV-COMP 2026	ULTIMATE GEMCUTTER	3rd place in the <i>Huawei Concurrency Challenge</i> (lead developer)
	ULTIMATE AUTOMIZER	1st place in <i>Overall</i> ranking (co-developer)
SV-COMP 2025	ULTIMATE GEMCUTTER	3rd place in <i>ConcurrencySafety</i> category (lead developer)
	ULTIMATE AUTOMIZER	1st place in <i>Overall</i> ranking (co-developer)
SV-COMP 2024	ULTIMATE GEMCUTTER	2nd place in <i>ConcurrencySafety</i> category (lead developer)
	ULTIMATE AUTOMIZER	1st place in <i>Overall</i> ranking (co-developer)

SV-COMP 2023

ULTIMATE GEMCUTTER **3rd place** in *ConcurrencySafety* category
(lead developer)
ULTIMATE AUTOMIZER **1st place** in *Overall* ranking
(co-developer)

SV-COMP 2022

ULTIMATE GEMCUTTER **1st place** in *NoDataRace* demo category
3rd place in *ConcurrencySafety* category
(lead developer)

OTHER INFORMATION

Languages

GERMAN First language
ENGLISH Fluent, written and spoken
FRENCH DELF B1
CHINESE YCT I, HSK II

Summer Schools

2023 41st Summer School Marktoberdorf on Safety and Security through Formal
Verification, Marktoberdorf, Germany
events.model.in.tum.de/mod23/
2020 14th Summer School on Modelling and Verification of Parallel Processes
(MOVEP), virtual
projects-verimag.imag.fr/movep2020/
2019 VMCAI Winter School, Lisbon, Portugal
vmcaischool19.tecnico.ulisboa.pt

Exchange Programs

CHINA 2013 Chinese Bridge summer camp Confucius Institute
CHINA 2010 Student Exchange High School of Beijing University
FRANCE Student Exchanges 2007/2008, 2009/2010 and 2010/2011

20th May 2026